

Comparitive Performance Report for tentacle-3-way-replica-balance-reads-off vs tentacle-3-way-replica-balance-reads-on vs direct-reads-3-way-replica-balance-reads- off-pr2-rerun vs direct-reads-3-way-replica-balance-reads- on-pr2-split-256k vs direct-reads-3-way-replica-balance-reads- on-run2

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Comparison summary for tentacle-3-way-replica-balance-reads-off vs tentacle-3-way-replica-balance-reads-on vs direct-reads-3-way-replica-balance-reads- off-pr2-rerun vs direct-reads-3-way-replica-balance-reads- on-pr2-split-256k vs direct-reads-3-way-replica-balance-reads- on-run2

Sequential									
Read	tentacle_3_way-replica-balance-reads-off	tentacle_3_way-replica-balance-reads-on	%change	direct-reads-3-way-replica-balance-reads-off-pr2-rerun	%change	direct-reads-3-way-replica-balance-reads-on-pr2-split-256k	%change	direct-reads-3-way-replica-balance-reads-on-run2	%change
4K	141831@0.4ms	125772@1.7	59%	86307@4.4	-39%	107124@3.6	-24%	107034@3.6	-25%
8K	127635@0.5ms	116564@1.8	70%	83059@4.6	-35%	105859@3.6	-17%	105672@3.6	-17%
16K	114277@3.4ms	1198850@1.9	74%	81033@4.7	-29%	103541@3.7	-9%	103105@3.7	-10%
32K	96131@4.0ms	54447@2.5	61%	75013@5.1	-22%	97721@3.9	2%	97390@3.9	1%
64K	4923@5.1ms	7547@3.3	53%	4229@5.9	-14%	5599@4.5	14%	5606@4.5	14%
128K	7869@6.4ms	10647@4.7	35%	7116@7.1	-10%	9064@5.5	15%	8767@5.7	11%
256K	11495@8.8ms	2908@7.8	12%	10785@9.3	-6%	12649@7.9	10%	11861@8.5	3%
384K	12087@16.7ms	18866@7.8	6%	11635@17.3	-4%	13252@15.2	10%	11613@5.4	-4%
512K	12738@21.1ms	18804@5.2	1%	12690@21.2	-0%	12098@22.2	-5%	11295@11.9	-11%
768K	12620@31.9ms	18380@8.1	-2%	12926@31.2	2%	11923@33.8	-6%	10795@9.3	-14%
1024K	12438@10.8ms	18189@5.5	-2%	13329@30.2	7%	12347@32.6	-1%	11363@5.9	-9%
2048K	11426@11.7ms	18251@13.1	-10%	12220@11.0	7%	12617@10.6	10%	11182@6.0	-2%
4096K	10350@25.9ms	18070@13.3	-3%	11286@23.8	9%	12067@9.7	17%	10778@9.3	4%

Random									
Read	tentacle_3_way-replica-balance-reads-off	tentacle_3_way-replica-balance-reads-on	%change	direct-reads-3-way-replica-balance-reads-off-pr2-rerun	%change	direct-reads-3-way-replica-balance-reads-on-pr2-split-256k	%change	direct-reads-3-way-replica-balance-reads-on-run2	%change
4K	153787@2.1ms	171696@2.2	12%	103584@3.7	-33%	107397@3.6	-30%	108657@3.5	-29%
8K	146094@2.6ms	161660@2.4	11%	99601@3.2	-32%	103967@3.7	-29%	104567@3.7	-28%
16K	137395@2.8ms	52226@2.1	11%	96017@4.0	-30%	99361@3.9	-28%	100039@3.8	-27%
32K	116730@2.7ms	183590@2.9	14%	88133@4.3	-24%	91621@4.2	-22%	91446@4.2	-22%
64K	6059@2.8ms	7059@3.0	17%	4998@5.0	-18%	5126@4.1	-15%	5127@4.9	-15%
128K	9360@3.6ms	10382@3.2	11%	8128@6.2	-13%	8416@6.0	-10%	8126@4.1	-13%
256K	12040@7.0ms	2336@2.7	2%	10455@8.0	-13%	11702@7.2	-3%	11012@3.0	-9%
384K	12176@4.1ms	2192@4.1	0%	12194@8.2	0%	12625@5.0	4%	11309@4.4	-7%
512K	12548@5.3ms	2491@5.4	-0%	12981@10.3	3%	11322@5.9	-10%	10471@6.4	-17%
768K	11838@4.2ms	2014@4.2	1%	13078@7.7	10%	11256@4.5	-5%	10410@4.8	-12%
1024K	12103@5.5ms	2103@5.5	0%	12788@10.5	6%	11979@5.6	-1%	11034@6.1	-9%

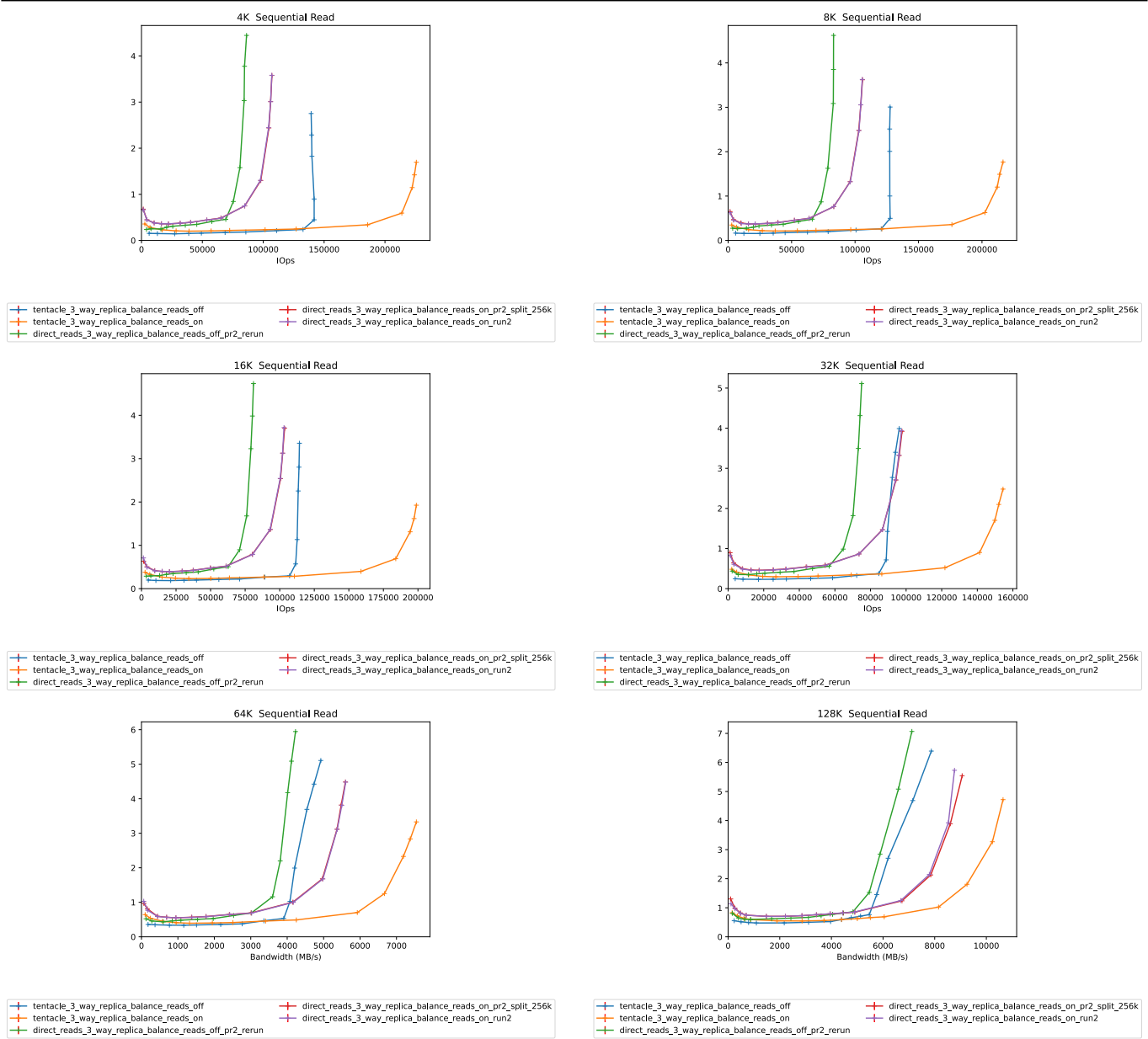
Comparison summary for tentacle-3-way-replica-balance-reads-off vs tentacle-3-way-replica-balance-reads-on vs direct-reads-3-way-replica-balance-reads-off-pr2-rerun vs direct-reads-3-way-replica-balance-reads-on-pr2-split-256k vs direct-reads-3-way-replica-balance-reads-on-run2

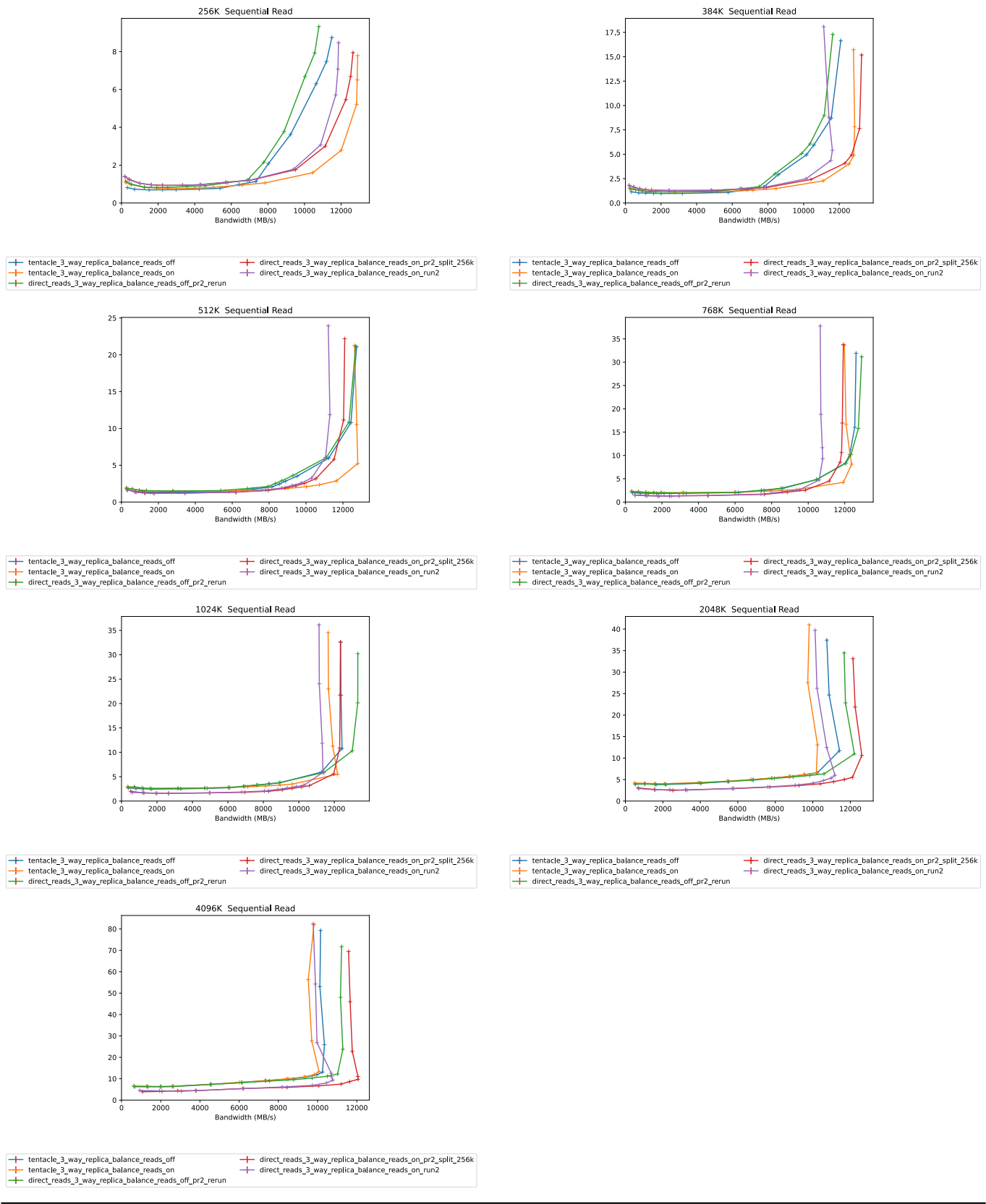
Random																				
Read	tentacle_3	tentacle3	way	change	direct	reads	%change	replicat	reads	%change	off	reads	%change	on	reads	%change	on	reads	%change	on
2048K	11155@12.0m	10678@12.5		-4%	12058@11.1		8%	12336@10.8		11%	11147@6.0		-0%							
4096K	10266@26.1m	10171@13.2		-1%	11218@23.9		9%	12088@11.1		18%	10898@9.2		6%							

Random																				
Read/Write	tentacle_3	tentacle3	way	change	direct	reads	%change	replicat	reads	%change	off	reads	%change	on	reads	%change	on	reads	%change	on
4K_70/30	82725@3.9ms	86455@4.4		5%	62792@6.1		-24%	63403@6.0		-23%	63744@6.0		-23%							
16K_70/30	75346@5.1ms	78340@3.3		4%	59158@6.5		-21%	59174@6.5		-21%	59743@6.4		-21%							
64K_30/70	2598@8.0ms	2662@9.4		2%	2252@11.1		-13%	2246@9.3		-14%	2246@9.3		-14%							
64K_70/30	3690@6.8ms	3886@5.4		5%	3127@5.3		-15%	3150@8.0		-15%	3147@6.6		-15%							

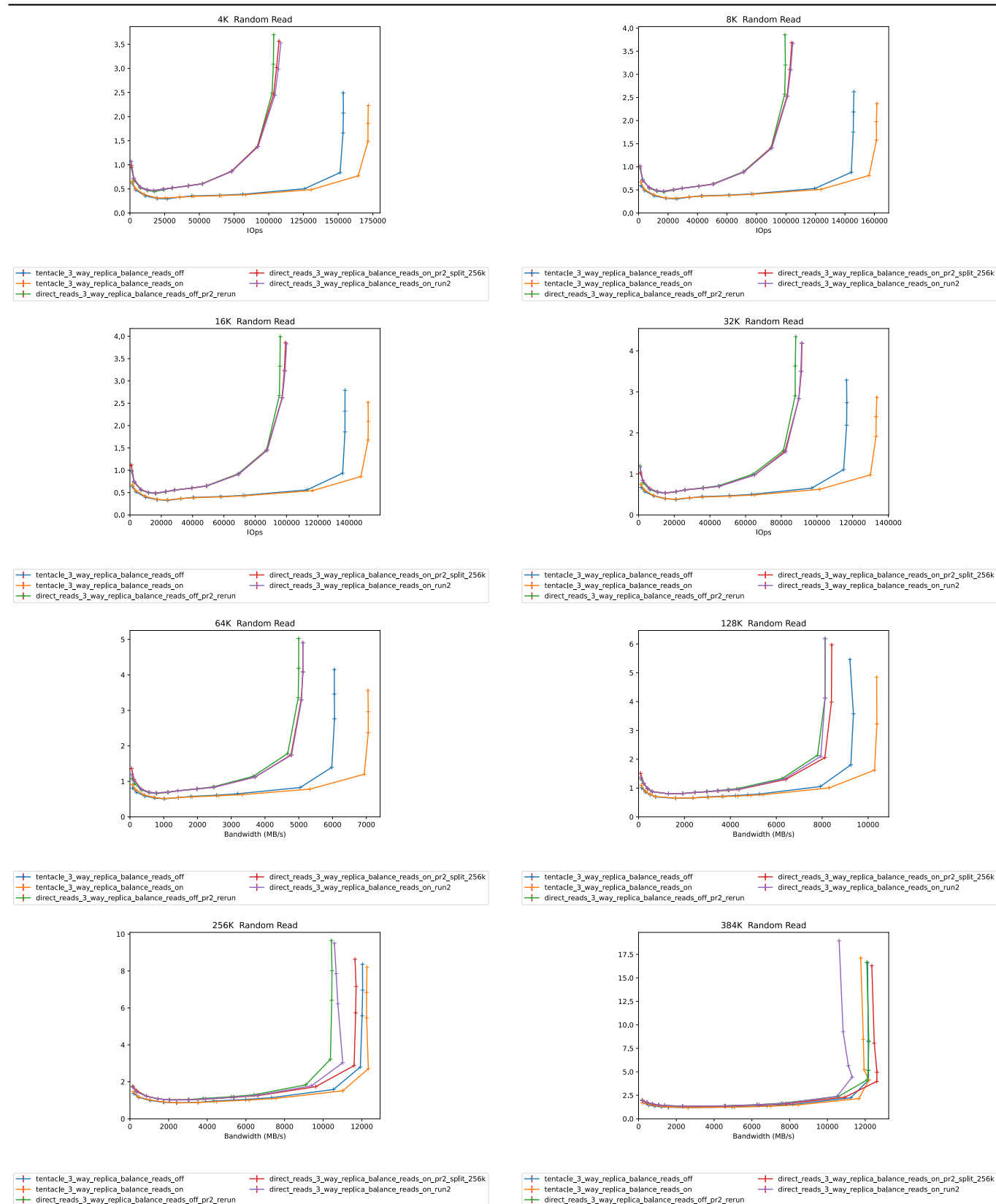
Response Curves

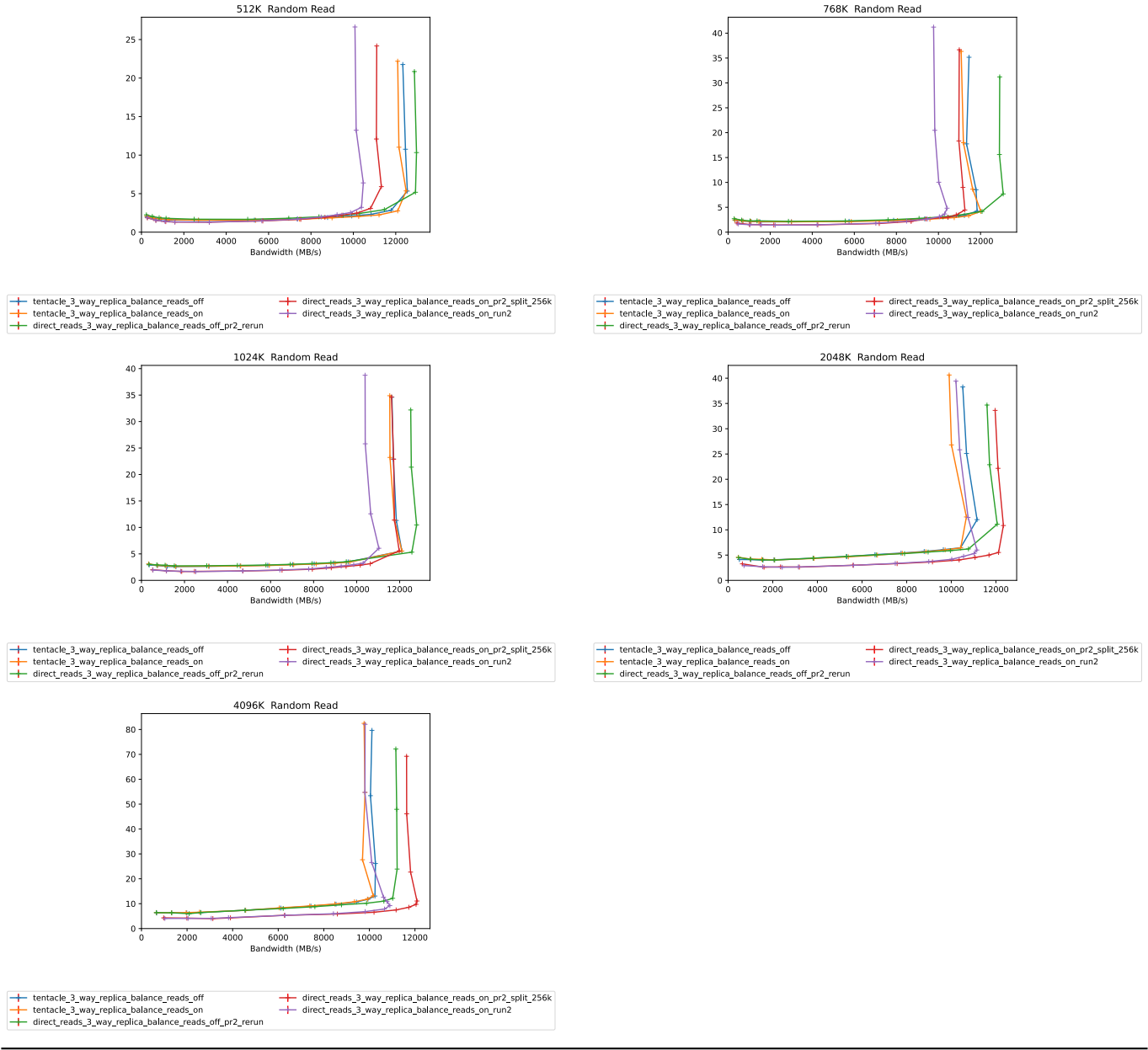
Sequential Read



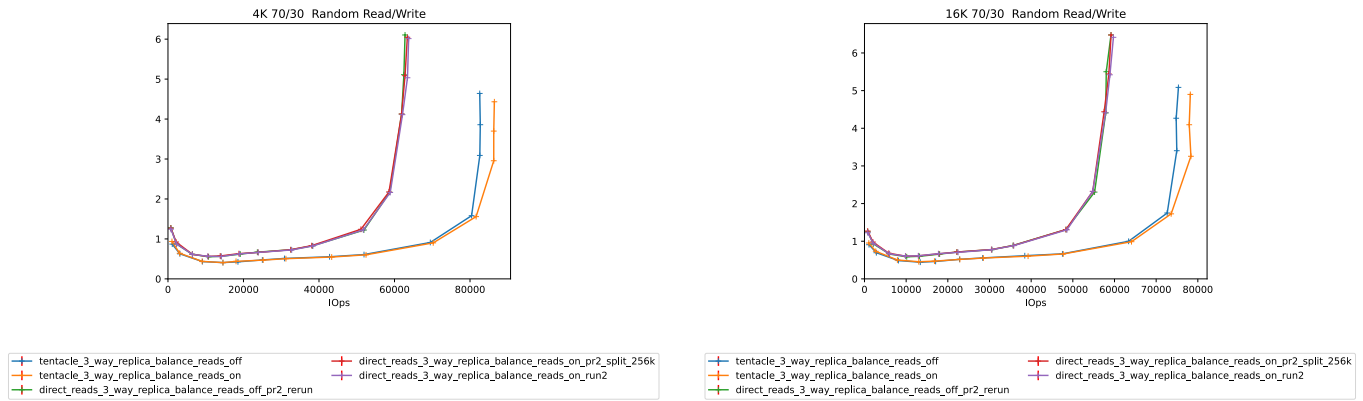


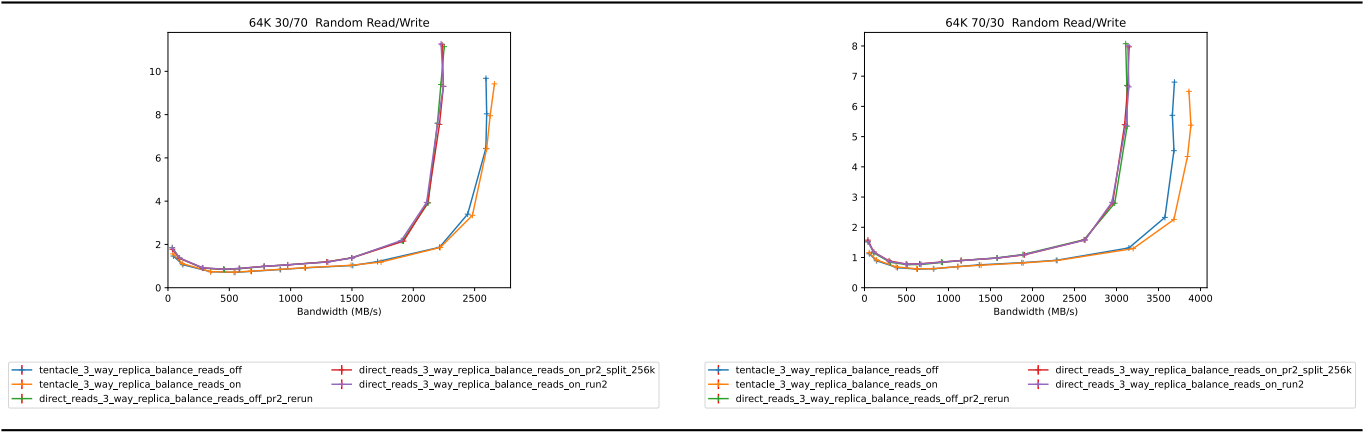
Random Read





Random Read/Write





Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384

16kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 16384
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

16krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 16384
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

16ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 16384
- total_iodepth:
- 1
- 2
- 4

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

256ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 262144
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

2Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

2Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 2097152
- total_iodepth:
- 1

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

384ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1
- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

4Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4096
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4096
- total_iodepth:

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
64kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 65536
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
64ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: cjames
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}
```